

# SANGRAMJEET GHOSH

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Hyderabad, Telangana, India

## Technical Skills

|                       |                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| Languages             | Python (3.10–3.13), TypeScript, SQL                                                                    |
| Backend & Cloud       | FastAPI, AsyncIO, Pydantic v2, Docker, PostgreSQL, AWS (EC2, S3, Lambda)                               |
| Agentic Frameworks    | LangGraph (StateGraph), LangChain, Model Context Protocol (MCP SDK), ReAct Agent Loops                 |
| LLM APIs & Inference  | Google Gemini Live (Bidirectional Audio), Groq (Llama-3), OpenAI API, Ollama (Local RAG), Transformers |
| Vector DBs & Memory   | PostgreSQL (pgvector Cosine Similarity), ChromaDB, SQLite State Machines, Knowledge Graphs             |
| Vision, Media & Audio | OpenCV (FER), GPU Speech Transcription, FFmpeg CLI, Librosa, PyAudio, Kokoro TTS, Playwright           |

## Professional Experience

**Freelance Software Engineer** Aug 2025 – Present  
*Independent* Remote

- Architected multimodal autonomous agents utilizing LangGraph state machines, Gemini Live streaming, and custom tool suites.
- Built Model Context Protocol (MCP) servers and persistent RAG pipelines with sub-10ms semantic retrieval across vector stores.

**Data Science & Backend Intern** Mar 2025 – Dec 2025  
*Inikola Technologies* Hyderabad, India

- Deployed FastAPI backend endpoints serving natural language scoring and feature extraction models under strict latency budgets.
- Built automated data validation pipelines using Pydantic and PostgreSQL to sanitize and index unstructured text feeds.

## Featured Projects

**Sebastian — Autonomous AI Operating System & Private Digital Butler** [github.com/JeetInTech](https://github.com/JeetInTech)  
*Python, Gemini Live API, Kokoro Neural TTS, PostgreSQL, pgvector, Model Context Protocol, PyAudio*

- Built a full-duplex, 25-module voice operating system via Gemini Live (~400ms turnaround) with barge-in interruption.
- Orchestrated 92 registered tools across Green/Yellow permission gates via `npx` child-process MCP servers (GitHub, Playwright, Filesystem).
- Engineered turn-by-turn semantic memory retrieval powered by **pgvector cosine similarity (sub-10ms)** and persistent Knowledge Graph.

**Genesis: Autonomous Multi-Board Job Application Agent** [github.com/JeetInTech](https://github.com/JeetInTech)  
*Python, Playwright, Groq LLM API, SQLite, BeautifulSoup4, Jinja2, pdflatex Engine*

- Developed an autonomous recruitment agent executing end-to-end applications across 5 job portals via Playwright and CDP.
- Parsed job listings with Groq Llama-3, dynamically generated tailored LaTeX resumes, and compiled PDFs via local `pdflatex` in 1.8s.
- Managed autonomous multi-step questionnaire answering and form filling with SQLite finite state tracking and screenshot audit trails.

**Virtual Therapist — LangGraph Autonomous Multi-Modal Agent** [github.com/JeetInTech](https://github.com/JeetInTech)  
*Python, LangGraph, ChromaDB, OpenCV (FER), Librosa, Groq LLM, Pydantic v2*

- Implemented a LangGraph StateGraph ReAct loop orchestrating **24 deterministic domain tools** and episodic conversational memory.
- Combined real-time facial micro-expression recognition (FER/OpenCV) and acoustic voice prosody analysis (Librosa) into an affective state vector.
- Built deterministic safety guardrails intercepting crisis indicators in <50ms via rule-based checks that run before any LLM call.

**Neuroviai Multi-Model LLM Routing & Gateway Engine** [neuroviai.dev](https://neuroviai.dev)  
*Python, FastAPI, Pydantic v2, Groq, Google Gemini Pro, HuggingFace, Local Ollama*

- Designed a 4-tier circuit-breaking LLM gateway (Gemini Pro → Groq → HuggingFace → Ollama) achieving 99.7% uptime under 12k tokens/min rate limits.
- Implemented exponential backoff with jitter on HTTP 429 and Pydantic v2 schema validation on model outputs.

**Genesis: Video Pipeline — Autonomous YouTube Shorts Agent** [github.com/JeetInTech](https://github.com/JeetInTech)  
*Python, Ollama, Groq LLM, Edge-TTS, FFmpeg CLI, YouTube Data API v3, OAuth 2.0*

- Built an autonomous agent on a daily schedule that generates scripts, synthesizes voiceovers, and composites video via FFmpeg with zero manual intervention.
- Sustained continuous unattended operation for 100+ days, autonomously publishing 138 Shorts via the YouTube Data API v3.

## Education

**Bachelor of Technology in Computer Science and Engineering (AI & ML)** Graduated May 2026  
Malla Reddy University, Hyderabad, Telangana **GPA: 8.0 / 10.0**